

Modeling Linear and Integer Programs

Part 2a: Linear Programs

IE 311 – Operations Research I

- An **extremely** important aspect of Operations Research is to *formulate* the optimization problem correctly.
- Another **extremely** important aspect of Operations Research is to *implement and solve* the formulated optimization problem.
- We will spend the first 5–6 weeks of the semester solely focusing on how to formulate Linear and Mixed-Integer Linear Programming problems and solve them using the Python–Gurobi interface.

Recall: Types of Linear Optimization Models

An optimization problem with linear relations between **continuous** variables is a **Linear Program**:

$$\max; 2x_1 + 5x_2 \quad \text{s.t.} \quad x_1 + 2x_2 \leq 5, ; ; x_1, x_2 \geq 0$$

An optimization problem with linear relations between **integer** variables is a **(Pure) Integer (Linear) Program**:

$$\max; 2x_1 + 5x_2 \quad \text{s.t.} \quad x_1 + 2x_2 \leq 5, ; ; x_1, x_2 \geq 0 \text{ and } x_1, x_2 \text{ integer}$$

An optimization problem with linear relations between **continuous and integer** variables is a **Mixed-Integer Linear Program**:

$$\max; 2x_1 + 5x_2 \quad \text{s.t.} \quad x_1 + 2x_2 \leq 5, ; ; x_1, x_2 \geq 0 \text{ and } x_1 \text{ integer}$$

A First Example: Farmer LP

- A farmer must determine how many acres of corn and wheat to plant.
- An acre of corn yields 10 bushels of corn and requires 4 hours of labor per week.
- An acre of wheat yields 25 bushels of wheat and requires 10 hours of labor per week.
- All corn can be sold at \$3 a bushel and all wheat can be sold at \$4 a bushel.
- Seven acres of land and 40 hours per week of labor are available.
- Government regulations require that at least 30 bushels of corn be produced.
- How can the farmer maximize the total revenue?

Parameters (what are given?)

- Type of grains (corn and wheat)
- Yield of each type (10 bushels/acre for corn, 25 bushels/acre for wheat)
- Revenue of each type (\$3/bushel for corn, \$4/bushel for wheat)
- Area availability (7 acres in total)
- Labor requirement (4 hrs/acre for corn, 10 hrs/acre for wheat)
- Labor availability (40 hrs in total)
- Government regulations (at least 30 bushels of corn)

Decision variables (what can be decided/changed?)

- c : amount of acres allocated to corn
- w : amount of acres allocated to wheat

Objective function (what is the aim?) – maximize the total revenue:

$$\max; 3 \cdot 10 c + 4 \cdot 25 w$$

Constraints (what should be obeyed/respected?)

$$c + w \leq 7 \quad (\text{land availability})$$

$$4c + 10w \leq 40 \quad (\text{labor availability})$$

$$10c \geq 30 \quad (\text{government regulations})$$

$$c, w \geq 0 \quad (\text{nonnegativity})$$

- **Proportionality and additivity** (linear relation between variables), e.g. $\text{labor used} = (\text{labor/acre for corn}) \cdot (\text{acres of corn}) + (\text{labor/acre for wheat}) \cdot (\text{acres of wheat})$. A nonlinear relation would violate this.
- **Divisibility** (continuity of variables), e.g. planting 3.0 acres of corn and 2.8 acres of wheat is possible. Integer variables would violate this.
- **Certainty** (deterministic parameters), e.g. it is certain that 1 acre of corn yields 10 bushels. Random/uncertain parameters would violate this.

Farmer LP: Simplest (but Dirty) Python–Gurobi Implementation

```
from gurobipy import GRB, Model
model = Model('Farmer')
corn = model.addVar(lb=0, ub=GRB.INFINITY, vtype=GRB.CONTINUOUS) # acres of corn
wheat = model.addVar(lb=0, ub=GRB.INFINITY, vtype=GRB.CONTINUOUS) # acres of wheat
model.setObjective(3*10*corn + 4*25*wheat, GRB.MAXIMIZE) # total revenue
model.addConstr(corn + wheat <= 7) # total acres available
model.addConstr(4*corn + 10*wheat <= 40) # total labor hours available
model.addConstr(10*corn >= 30) # government requirement
model.optimize() # optimize the model
```

- Giapetto's, Inc. manufactures two types of wooden toys: soldiers and trains.
- Each **soldier** built sells for \$27, uses \$10 of raw materials, increases variable labor/overhead costs by \$14, and requires 2 hours of finishing and 1 hour of carpentry labor.
- Each **train** built sells for \$21, uses \$9 of raw materials, increases variable labor/overhead costs by \$10, and requires 1 hour of finishing and 1 hour of carpentry labor.
- Each week Giapetto has only 100 finishing hours and 80 carpentry hours.
- Demand for trains is unlimited, but at most 40 soldiers are bought each week.
- Giapetto wants to maximize weekly profit (revenues – costs).

Decision variables: s = number of soldiers built, t = number of trains built.

Objective function – maximize weekly profit:

$$\max; [27s + 21t] - [(10 + 14)s + (9 + 10)t]$$

Constraints (what should be obeyed/respected?)

$$2s + t \leq 100 \quad (\text{finishing hours})$$

$$s + t \leq 80 \quad (\text{carpentry hours})$$

$$s \leq 40 \quad (\text{demand for soldiers})$$

$$s, t \geq 0 \quad (\text{nonnegativity})$$

Is the divisibility assumption satisfied?

Two Python–Gurobi implementations of the Giapetto LP are uploaded to SUCourse:

- **Implementation 1** (ModelHardcoded): all parameters are *hard-coded*. Not the best practice – one must edit the function whenever a parameter changes.
- **Implementation 2** (ModelBetterImplemented): all parameters are passed as *arguments*. A better practice – one only changes the arguments.

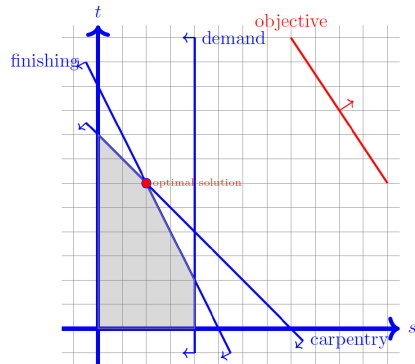
Both implementations are accepted when grading, but Implementation 2 is strongly suggested whenever possible.

- Both examples so far (Farmer and Giapetto) have only two variables.
- LPs with two variables can be solved by the **Graphical Solution Method**.
- This gives valuable intuition about the possible outcomes of LPs:
- LP is **infeasible** (no feasible solution exists)
- LP is **feasible** (a feasible solution exists)
 - unbounded objective, or
 - bounded objective (unique optimal solution, or multiple optimal solutions)
- The Simplex Method (later) is based on the intuition from the graphical method.

Graphical Solution Method: Giapetto LP

$$\begin{array}{ll}\max & 3s + 2t \\ \text{s.t.} & 2s + t \leq 100 \\ & s + t \leq 80 \\ & s \leq 40 \\ & s, t \geq 0\end{array}$$

(profit)
(finishing)
(carpentry)
(demand)
(nonnegativity)



- A post office requires different numbers of full-time employees on different days of the week: 17, 13, 15, 19, 14, 16, 11 (Monday through Sunday).
- Union rules state that each full-time employee must work five consecutive days and then receive two days off (e.g. Monday–Friday on, Saturday–Sunday off).
- The post office wants to meet its daily requirements using only full-time employees.
- Formulate an LP and IP to minimize the number of full-time employees hired.

Define

x_i = number of employees who begin their five working days on day i , $i = 1, \dots, 7$.

Each such employee works days $i, i + 1, \dots, i + 4$ (cyclically through the week) and is then off for two days. Hence each daily requirement is covered by the five "shifts" that include that day.

$$\begin{array}{ll}
 \min & x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 \quad (\text{total employees}) \\
 \text{s.t.} & x_1 + x_4 + x_5 + x_6 + x_7 \geq 17 \quad (\text{day 1}) \\
 & x_1 + x_2 + x_5 + x_6 + x_7 \geq 13 \quad (\text{day 2}) \\
 & x_1 + x_2 + x_3 + x_6 + x_7 \geq 15 \quad (\text{day 3}) \\
 & x_1 + x_2 + x_3 + x_4 + x_7 \geq 19 \quad (\text{day 4}) \\
 & x_1 + x_2 + x_3 + x_4 + x_5 \geq 14 \quad (\text{day 5}) \\
 & x_2 + x_3 + x_4 + x_5 + x_6 \geq 16 \quad (\text{day 6}) \\
 & x_3 + x_4 + x_5 + x_6 + x_7 \geq 11 \quad (\text{day 7}) \\
 & x_1, \dots, x_7 \geq 0 \quad (\text{LP})
 \end{array}$$

For the **IP**, additionally require $x_1, \dots, x_7 \geq 0$ **and integer**.

- Dorian Auto manufactures luxury cars and trucks; likely customers are high-income women and men.
- It buys 1-minute commercial spots on comedy shows and football games.
- Each comedy ad is seen by 7M high-income women and 2M high-income men.
- Each football ad is seen by 2M high-income women and 12M high-income men.
- A comedy ad costs \$50,000; a football ad costs \$100,000.
- Dorian wants the ads seen by at least 28M high-income women and 24M high-income men.
- (a) Model as an LP and solve in Python–Gurobi.
- (b) Suppose radio ads are also considered (\$25,000 each, reaching 3M women and 4M men). Re-run the model.

My diet must come from four food groups (chocolate cake, ice cream, soda, cheesecake), available as: brownies, chocolate ice cream, cola, pineapple cheesecake (costing 50¢, 20¢, 30¢, 80¢ respectively). Each day I must ingest ≥ 500 calories, ≥ 6 oz chocolate, ≥ 10 oz sugar, ≥ 8 oz fat.

Ingredient	Brownie	Choc. ice cream	Cola	Pineapple cheesecake
Calories	400	200	150	500
Chocolate (oz)	3	2	0	0
Sugar (oz)	2	2	4	4
Fat (oz)	2	4	1	5

Formulate an LP to meet the requirements at minimum cost, and generalize to m ingredients and n food types.

Let x_1, x_2, x_3, x_4 be the amounts of brownie, chocolate ice cream, cola, and pineapple cheesecake. **Objective** – minimize cost:

$$\min; 50x_1 + 20x_2 + 30x_3 + 80x_4$$

Diet LP – Specific Formulation (Constraints)

$$\begin{aligned} 400x_1 + 200x_2 + 150x_3 + 500x_4 &\geq 500 && \text{(calories)} \\ 3x_1 + 2x_2 + 0x_3 + 0x_4 &\geq 6 && \text{(chocolate)} \\ 2x_1 + 2x_2 + 4x_3 + 4x_4 &\geq 10 && \text{(sugar)} \\ 2x_1 + 4x_2 + 1x_3 + 5x_4 &\geq 8 && \text{(fat)} \\ x_1, x_2, x_3, x_4 &\geq 0 && \text{(nonnegativity)} \end{aligned}$$

Parameters: a_{ij} = amount of ingredient i in food j ; b_i = requirement for ingredient i ; c_j = cost of food j ($i = 1, \dots, m$, $j = 1, \dots, n$). **Decision variables:** x_j = amount of food j consumed.

$$\begin{aligned} \min \quad & \sum_{j=1}^n c_j x_j && \text{(total cost)} \\ \text{s.t.} \quad & \sum_{j=1}^n a_{ij} x_j \geq b_i, \quad i = 1, \dots, m && \text{(requirement } i) \\ & x_j \geq 0, \quad j = 1, \dots, n && \text{(nonnegativity)} \end{aligned}$$

A company produces medical diagnostic equipment at two factories; three medical centers placed orders this month. Decide how many units to ship from each factory to each customer. Formulate as an LP.

	Customer 1	Customer 2	Customer 3	Output
Factory 1	\$600/item	\$800/item	\$700/item	400 units
Factory 2	\$400/item	\$900/item	\$600/item	500 units
Order size	300 units	200 units	400 units	

Decision variables: we need to know how many items are sent from each factory to each customer.

Constraints: we must obey the factory capacities and satisfy customer demands.

- Question: what is the total amount sent from Factory 1?
- Question: what is the total amount received by Customer 2?

Transportation LP: Specific Model

x_{ij} = units shipped from factory i to customer j ($i = 1, 2, j = 1, 2, 3$).

$$\begin{array}{ll} \min & 600x_{11} + 800x_{12} + 700x_{13} + 400x_{21} + 900x_{22} + 600x_{23} \quad (\text{cost}) \\ \text{s.t.} & x_{11} + x_{12} + x_{13} \leq 400 \quad (\text{factory 1}) \\ & x_{21} + x_{22} + x_{23} \leq 500 \quad (\text{factory 2}) \\ & x_{11} + x_{21} \geq 300 \quad (\text{customer 1}) \\ & x_{12} + x_{22} \geq 200 \quad (\text{customer 2}) \\ & x_{13} + x_{23} \geq 400 \quad (\text{customer 3}) \\ & x_{ij} \geq 0 \quad (\text{nonnegativity}) \end{array}$$

Transportation LP: General Model

Parameters: c_{ij} = unit cost factory $i \rightarrow$ customer j ; s_i = output of factory i ; d_j = demand of customer j ($i = 1, \dots, m, j = 1, \dots, n$). **Variables:** x_{ij} = units shipped.

$$\begin{aligned} \min \quad & \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} && \text{(total cost)} \\ \text{s.t.} \quad & \sum_{j=1}^n x_{ij} \leq s_i, \quad i = 1, \dots, m && \text{(capacity } i) \\ & \sum_{i=1}^m x_{ij} \geq d_j, \quad j = 1, \dots, n && \text{(demand } j) \\ & x_{ij} \geq 0 && \text{(nonnegativity)} \end{aligned}$$

Sunco Oil makes three types of gasoline by blending three types of crude oil. Sunco can buy up to 5,000 barrels of each crude daily. Octane/sulfur requirements: gas 1 octane ≥ 10 , sulfur $\leq 1\%$; gas 2 octane ≥ 8 , sulfur $\leq 2\%$; gas 3 octane ≥ 6 , sulfur $\leq 1\%$. Transforming oil into gas costs \$4/barrel; refinery capacity is 14,000 barrels/day. Daily demand: gas 1 = 3,000, gas 2 = 2,000, gas 3 = 1,000 barrels. Each \$1/day on advertising a gas raises its demand by 10 barrels.

Gas	Price(\$)	Oil	Cost(\$)	Octane	Sulfur(%)
1	70	1	45	12	0.5
2	60	2	35	6	2.0
3	50	3	25	8	3.0

Decision variables: how much oil of each type is used in each gas blend, and the advertising amount for each gas.

Constraints: obey crude oil availability and satisfy gas demands; gas blends must satisfy the quality (octane/sulfur) requirements.

- Question: total amount of oil 1 used? Total amount of gas 3 blended?
- Question: average octane rating of gas 3 blended?

Blending LP: Decision Variables and Objective

x_{ij} = amount of crude oil i used in gas j ; y_j = money spent advertising gas j ($i, j = 1, 2, 3$).

$$\begin{aligned} \max \quad & 70 \sum_{i=1}^3 x_{i1} + 60 \sum_{i=1}^3 x_{i2} + 50 \sum_{i=1}^3 x_{i3} \\ & - 45 \sum_{j=1}^3 x_{1j} - 35 \sum_{j=1}^3 x_{2j} - 25 \sum_{j=1}^3 x_{3j} - 4 \sum_{i=1}^3 \sum_{j=1}^3 x_{ij} - \sum_{j=1}^3 y_j \end{aligned}$$

(gas revenues – oil costs – processing cost – advertising cost)

$$\sum_{j=1}^3 x_{ij} \leq 5000, \quad i = 1, 2, 3 \quad (\text{oil availability})$$

$$\sum_{i=1}^3 x_{i1} = 3000 + 10y_1, \quad \sum_{i=1}^3 x_{i2} = 2000 + 10y_2, \quad \sum_{i=1}^3 x_{i3} = 1000 + 10y_3 \quad (\text{demand})$$

$$\sum_{i=1}^3 \sum_{j=1}^3 x_{ij} \leq 14000 \quad (\text{capacity})$$

$$12x_{1j} + 6x_{2j} + 8x_{3j} \geq r_j(x_{1j} + x_{2j} + x_{3j}), \quad r = (10, 8, 6) \quad (\text{octane})$$

$$0.5x_{1j} + 2x_{2j} + 3x_{3j} \leq \sigma_j(x_{1j} + x_{2j} + x_{3j}), \quad \sigma = (1, 2, 1) \quad (\text{sulfur})$$

$$x_{ij}, y_j \geq 0 \quad (\text{nonnegativity})$$

Powerco has three plants supplying four cities. Decide how many Mwh to send from each plant to each city; formulate, implement and solve as an LP. Then add the requirement of at most 25 Mwh from any plant to any city and re-run.

	City 1	City 2	City 3	City 4	Capacity
Plant 1	\$8	\$6	\$10	\$9	≤ 35 Mwh
Plant 2	\$9	\$12	\$13	\$7	≤ 50 Mwh
Plant 3	\$14	\$9	\$16	\$5	≤ 45 Mwh
Demand	$= 45$	$= 20$	$= 30$	$= 30$	

(costs in \$/kwh; demands in Mwh)

Rylon makes Brute and Chanelle perfumes. Raw material costs \$3/lb and 1 lb needs 1 hour of lab time, yielding 3 oz Regular Brute (\$7/oz) and 4 oz Regular Chanelle (\$6/oz). Regular Brute can be further processed into Luxury Brute (\$18/oz; +3 lab hrs and \$4 per oz) and Regular Chanelle into Luxury Chanelle (\$14/oz; +2 lab hrs and \$4 per oz). Each year: 6,000 lab hours, up to 4,000 lb of raw material. Maximize profit (lab cost is fixed).

m = raw material (lb); B_r, C_r = oz of regular Brute/Chanelle produced; B_l, C_l = oz of luxury Brute/Chanelle produced.

$$\begin{array}{ll}\max & 7B_r + 6C_r + (18 - 4)B_l + (14 - 4)C_l - 3m \quad (\text{profit}) \\ \text{s.t.} & m + 3B_l + 2C_l \leq 6000 \quad (\text{lab hours}) \\ & B_r + B_l = 3m \quad (\text{Brute yield}) \\ & C_r + C_l = 4m \quad (\text{Chanelle yield}) \\ & m \leq 4000 \quad (\text{availability}) \\ & m, B_r, C_r, B_l, C_l \geq 0 \quad (\text{nonnegativity})\end{array}$$

We now model a multi-period production/inventory planning problem: in each period we choose how much to produce, paying production and inventory-holding costs while meeting demand on time.

Decision variables: the production and ending inventory of each month.

Constraints: obey the inventory (material balance) requirements linking consecutive months.

x_t = production in month t ($t = 1, \dots, 4$); l_t = inventory at end of month t ($t = 0, \dots, 4$).

$$\begin{array}{ll} \min & 20x_1 + 15x_2 + 35x_3 + 100x_4 + 15l_1 + 25l_2 + 10l_3 \quad (\text{total cost}) \\ \text{s.t.} & l_0 + x_1 = l_1 + 40 \quad (\text{month 1}) \\ & l_1 + x_2 = l_2 + 70 \quad (\text{month 2}) \\ & l_2 + x_3 = l_3 + 30 \quad (\text{month 3}) \\ & l_3 + x_4 = l_4 + 50 \quad (\text{month 4}) \\ & l_0 = 10, \quad l_4 = 0, \quad x_t, l_t \geq 0 \end{array}$$

A Few "Tricks" to Linearize Special Nonlinear Expressions

Some nonlinear expressions can be reformulated as linear programs, for example absolute values, the minimum/maximum of linear functions, and concave piecewise-linear objectives. The next examples illustrate these tricks.

Machine Balancing Example

A machine shop has a drill press and a milling machine producing two parts A and B . Required time (minutes) per unit is shown below. The shop must produce at least 50 units total and at least 30 units of A , and at most 100 of A and 100 of B . Formulate an LP to minimize the absolute difference between the total running time of the drill press and that of the milling machine.

Part	Drill Press	Milling Machine
A	3	4
B	5	3

a, b = numbers of part A, B produced; z = absolute difference of machine running times. Note

$$z = |(3a + 5b) - (4a + 3b)| = |2b - a|.$$

$$\begin{array}{ll}\min & z \\ \text{s.t.} & z \geq 2b - a, \quad z \geq a - 2b \\ & a + b \geq 50 \\ & 30 \leq a \leq 100, \quad 0 \leq b \leq 100\end{array}$$

Piecewise Linear Example

A farm grows two mango types (yields: type 1 = 1000 units, type 2 = 700 units); all are sold, either to a retail company or in the local market. Retail contract: at least 60% of mangoes sold to retail must be type 2; demand is 1000 units at \$3/unit (need not be met exactly); a \$0.05/unit penalty applies for each unit short of 1000; each unit beyond the first 1200 is bought at \$2.5/unit. Local market prices: type 1 = \$2.2/unit, type 2 = \$3.1/unit (any amount sellable). Formulate an LP to maximize net revenue.

Piecewise Linear Example: Variables, Constraints, Objective

R_i = type i mangoes sold to retail; T = total sold to retail; z = revenue from the retailer; L_i = type i sold locally ($i = 1, 2$).

$$L_1 + R_1 = 1000 \quad (\text{type 1})$$

$$L_2 + R_2 = 700 \quad (\text{type 2})$$

$$R_1 + R_2 = T \quad (\text{total to retailer})$$

$$R_2 \geq 0.6 T \quad (\text{type-2 ratio})$$

$$R_1, R_2, T, L_1, L_2 \geq 0 \quad (\text{nonnegativity})$$

Objective: $\max; 2.2L_1 + 3.1L_2 + z$.

The revenue z obtained from the retailer is the piecewise-linear function

$$z = \begin{cases} 3T - 0.05(1000 - T) & 0 \leq T \leq 1000 \\ 3T & 1000 \leq T \leq 1200 \\ 3 \cdot 1200 + 2.5(T - 1200) & T \geq 1200 \end{cases}$$

Piecewise Linear Example: Concave PWL is "Easy"

Since the function above is **concave** piecewise linear,

$$z = \min 3T - 0.05(1000 - T), \quad 3T, \quad 3 \cdot 1200 + 2.5(T - 1200).$$

Maximizing it is therefore equivalent to the linear program

$$\max; z \quad \text{s.t.} \quad z \leq 3T - 0.05(1000 - T), \quad z \leq 3T, \quad z \leq 3 \cdot 1200 + 2.5(T - 1200).$$

This is an LP!

Piecewise Linear Example: Complete Model

$$\begin{aligned} \max \quad & 2.2L_1 + 3.1L_2 + z \\ \text{s.t.} \quad & L_1 + R_1 = 1000, \quad L_2 + R_2 = 700 \\ & R_1 + R_2 = T, \quad R_2 \geq 0.6 T \\ & 3T - 0.05(1000 - T) \geq z \quad (\text{piece 1}) \\ & 3T \geq z \quad (\text{piece 2}) \\ & 3 \cdot 1200 + 2.5(T - 1200) \geq z \quad (\text{piece 3}) \\ & R_1, R_2, T, L_1, L_2 \geq 0 \end{aligned}$$

Single Facility Rectilinear Distance Location Problem

A warehouse serves six plants with given coordinates and demands:

x-coordinate	y-coordinate	Demand
1	1	50
5	12	60
2	8	19
4	4	20
8	6	32
10	7	23

Parameters: (a_i, b_i) = coordinates of plant i ; d_i = demand of plant i . **Variables:** (x, y) = warehouse location; (u_i, v_i) = axis distances, with $u_i = |x - a_i|$, $v_i = |y - b_i|$ ($i = 1, \dots, 6$).

$$\begin{aligned} \min \quad & \sum_{i=1}^6 d_i(u_i + v_i) && \text{(total cost)} \\ \text{s.t.} \quad & u_i \geq x - a_i, \quad u_i \geq a_i - x, \quad i = 1, \dots, 6 && \text{(horizontal)} \\ & v_i \geq y - b_i, \quad v_i \geq b_i - y, \quad i = 1, \dots, 6 && \text{(vertical)} \\ & u_i, v_i \geq 0 && \text{(nonnegativity)} \end{aligned}$$

Sailco must decide how many sailboats to produce over four quarters. Demands: 40, 60, 75, 25. Demands must be met on time; initial inventory is 10 boats. Each quarter up to 40 boats can be made with regular-time labor at \$400 each, plus additional boats with overtime labor at \$450 each. A holding cost of \$20/boat is incurred on end-of-quarter inventory. Minimize total production plus inventory cost.

Sailco Inventory: Model

r_t = regular-time production, o_t = overtime production in quarter t ($t = 1, \dots, 4$); l_t = end-of-quarter inventory ($t = 0, \dots, 4$).

$$\begin{aligned} \min \quad & 400 \sum_{t=1}^4 r_t + 450 \sum_{t=1}^4 o_t + 20 \sum_{t=1}^4 l_t && \text{(total cost)} \\ \text{s.t.} \quad & l_0 + r_1 + o_1 = l_1 + 40 && \text{(quarter 1)} \\ & l_1 + r_2 + o_2 = l_2 + 60 && \text{(quarter 2)} \\ & l_2 + r_3 + o_3 = l_3 + 75 && \text{(quarter 3)} \\ & l_3 + r_4 + o_4 = l_4 + 25 && \text{(quarter 4)} \\ & l_0 = 10, \ 0 \leq r_t \leq 40, \ o_t \geq 0, \ l_t \geq 0 \end{aligned}$$